F18CE2 - MV Calculus and Real Analysis B: Tutorial 7

1 State Rolle's Theorem. Use Rolle's Theorem to prove that for any $\gamma \in \mathbb{R}$, the equation

$$x^5 - 5\,x + \gamma = 0$$

does not have two solutions greater than 1.

2 Use the Intermediate Value Theorem and Rolle's Theorem to prove that the equation $x + e^x = 2$ has exactly one solution in $\mathbb{R}$.

3 Use the Intermediate Value Theorem and Rolle's Theorem to prove that the equation

$$x^5 + x^3 + x = 1$$

has exactly one solution between -1 and 1.

4 Use the Intermediate Value Theorem and Rolle's Theorem to prove that the equation

$$x \exp(\sin x) = \cos x$$

has exactly one solution between 0 and $\frac{\pi}{2}$.

5 Use the mean value theorems to prove the following inequalities, for all $x \geqslant 0$.

Hint: in each part of the question define a function $f : [0, \infty) \to \mathbb{R}$ such that f is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$, and for which the inequality is equivalent to $f(x) \geqslant 0$, for $x \geqslant 0$.

For arbitrary $x > 0$ the MVT shows that

$$f(x) = f(0) + x\,f'(c), \quad \text{for some } c \in (0, x).$$

So, if $f(0) \geqslant 0$ and $f' \geqslant 0$ on $[0, x]$, then $f \geqslant 0$ on $[0, x]$, which proves the result.

(a) $x \geqslant 1 - \cos x$.

(b) $\ln(1 + x) \geqslant x - \dfrac{x^2}{2}$.

(c) $(1+x)^\alpha \geqslant 1 + x^\alpha$, for any $\alpha \geqslant 1$. Deduce that if $a,\ b \geqslant 0$ then $(a+b)^\alpha \geqslant a^\alpha + b^\alpha$.

(d) $(1 + x)^n \geqslant 1 + nx$, for any integer $n \geqslant 1$.

(e) $(1 + x)^n \geqslant 1 + nx + \dfrac{1}{2}n(n-1)x^2$, for any integer $n \geqslant 1$
(use the second MVT here , i.e., the nth MVT with $n = 2$).
Verify that these inequalities are actually equalities when $n = 1,\ 2$.

6 (a) Define a sequence (a_n) by $a_n = 2^{1/n}, \quad n \geqslant 1$.
Use the inequality in Q. **5** 4 to prove that $\lim\limits_{n\to\infty} a_n = 1$.
[Hint: let $a_n = 1 + b_n$, with $b_n \geqslant 0$, and use the preceding inequality to get an inequality for b_n.]

(b) Define a sequence (a_n) by $a_n = n^{1/n}, \quad n \geqslant 1$.
Use the inequality in Q. **5** 5 to prove that $\lim\limits_{n\to\infty} a_n = 1$.
[Hint: follow the method of part (a).]

7 By applying the first MVT to the function $f(x) = x \ln x$, prove that for any integer $r \geqslant 1$,

$$(r+1)\ln(r+1) - r\ln r \leqslant 1 + \ln(r+1)\,.$$

By summing these inequalities from $r = 1$ to $r = n-1$, deduce that for any integer $n \geqslant 2$,

$$1 + n\,\ln n - n \leqslant \ln(n!)\,.$$

8 By applying the first MVT to the function $f(x) = \sqrt{x}$, prove that for any integer $n \geqslant 1$,

$$\frac{1}{2}\,(n+1)^{-1/2} \;\leqslant\; \sqrt{n+1} - \sqrt{n} \leqslant \frac{1}{2}\,n^{-1/2}\,.$$

Define a sequence (a_n) by

$$a_n = 1 + 2^{-1/2} + 3^{-1/2} + \cdots + n^{-1/2} - 2\sqrt{n}.$$

Use the preceding inequality to prove that (a_n) is decreasing.